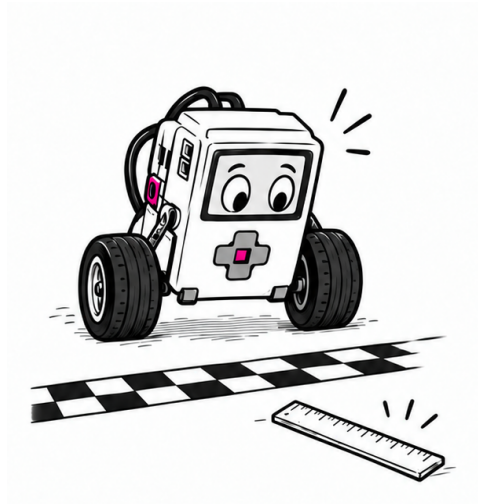


LOUDOUN ROBOTICS

Engineering Notebook

Think it. Try it. Make it better.



Robotics Foundations · Fall 2026
Stone Hill Middle School

YOUR NAME

TEAM / TEAMMATES

Your twelve-week adventure

One EV3 kit. Two builds. Twelve weeks of discovery.

Week	Date	Challenge	Pages
1	Sep 16	Build and drive	6-7
2	Sep 23	Find your grip	8-9
3	Sep 30	Speed or shove?	10-11
4	Oct 7	Choose your fighter	12-13
5	Oct 14	Make it move	14-15
6	Oct 21	Repeat and turn	16-17
7	Oct 28	Feel a hit	18-19
8	Nov 4	Can it stay in?	20-21
9	Nov 11	Find the opponent	22-23
10	Nov 18	Choose what happens	24-25
11	Dec 2	Make one improvement	26-27
12	Dec 9	Tournament day	28-29

Wednesdays, 6-7 PM. No session November 25. Your notebook stays with the team kit between sessions; take it home after Week 12.

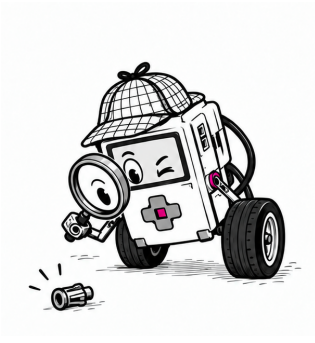
50 minutes planned + 10 minutes for late starts, batteries, or troubleshooting.

5 arrival · 7 idea/predict · 25 build/test · 8 record/share · 5 save/clean

Where to find help

Looking for...	Page
Sumo rules	30-31
Reference cards	32-33
First EV3 program	5
Debugging help	34
Skills checklist	35
Catch-up guide	36

Your notebook is a detective tool



A surprising result is a clue.

Predict first. Keep the numbers. Sketch what changed.

Engineers design, test, and improve things to meet a need within limits.

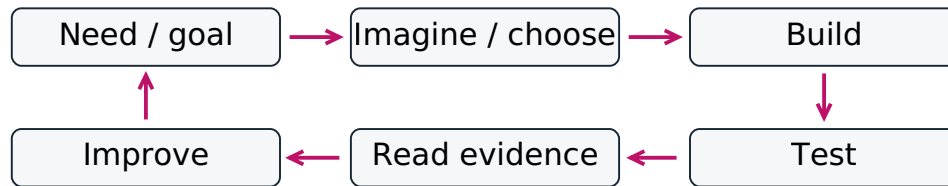
This is **your** book. Share team data; explain it in your own words.

A useful entry	Example - made-up data
Prediction	"I think it will travel 100 cm."
Setup	Same floor, wheels, power, start line, and program.
Three runs	96 cm · 97 cm · 98 cm
What I noticed	All three stopped short. Mean 97 cm; range 2 cm.
Next change	"Try slightly more motor rotation; measure again."

Show your thinking with words, numbers, labeled sketches, diagrams, flowcharts/code, or an explanation to a mentor when appropriate. Ask the mentor to help capture the key idea here. Keep earlier attempts visible; cross out a mistake once and add the correction.

A question I want to investigate

Everyone gets a turn



Start with what you know

What do we know for sure? What are we assuming?

What can we measure? What is the simplest test?

What does the evidence actually show?

Week	Builder	Programmer	Tester
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

Before a run

Clear the floor lane.

Say "testing."

Know the Back-button stop.

Before an adjustment

Stop the program.

Keep fingers and hair clear.

Ask a mentor about stalls or loose parts.

Rotate lead roles weekly; swap hands-on turns after each build section or test. Even when you are not the lead, be ready to explain what the team tested and what the evidence showed.

I can stop our robot. Name / date / mentor initials

Bevel Bot: your first program

One motor. One connected program. Make it move.

Build Bevel Bot first (Week 1, p. 6). Open **LEGO EV3 Classroom** and create a project.

- 1 Connect the Bevel Bot's medium motor to **output A**; connect the brick to the laptop by USB. Select your brick.
- 2 Use **single-motor** blocks for A. Build the stack below. Save as Team_Week01_Bevel_v01.
- 3 With wheels supported, ask a mentor to check the direction. Stop, place it in a clear lane, announce the test, then run.

When program starts

Set motor A speed to 20%

Wait 1 second

Run motor A for 1 rotation in the tested forward direction

Stop motor A; brake

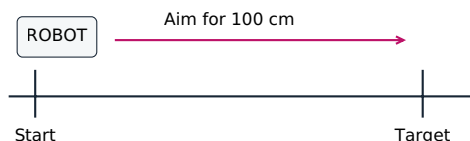
Use the brick's Back button to stop. One motor turn is not a promise of 100 cm. Measure the travel from the same front edge.

Week 4 changes the build: two independent drive motors on B/C. From Week 4 onward, use Movement blocks for that pair; see Programming Guide p. 2. Do not reuse the A-only program for the new base.

Build and drive



Mission: Build Bevel Bot. Make it drive and stop.



Key idea

Electric motor — A device that converts electrical energy into motion. Example: The EV3 motor turns an axle.

Program — A set of instructions a computer executes. Example: Run motor A, then stop.

Sequence — The order in which instructions run. Example: Wait first; move next.

Predict first

Where will the first run stop?

Try it

- 1 Build from [LEGO's Bevel Bot pictures](#). Take turns; have a mentor check wheels, gears, and cables.
- 2 Build the A-motor program (p. 5). Drive, stop, and measure from the same front edge.
- 3 Optional: tune toward 100 cm; record three runs with unchanged settings.

Optional: three runs toward 100 cm

Trial	Actual (cm)	Error: actual - 100
1		
2		
3		
Mean		
Range		

Same-sign errors may suggest systematic error (bias); scatter suggests random error. Three trials are clues, not proof.

Same-sign errors, scattered errors, or both?

Beyond sumo • Where else would a robot need to stop accurately?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Wheel diameter (cm)

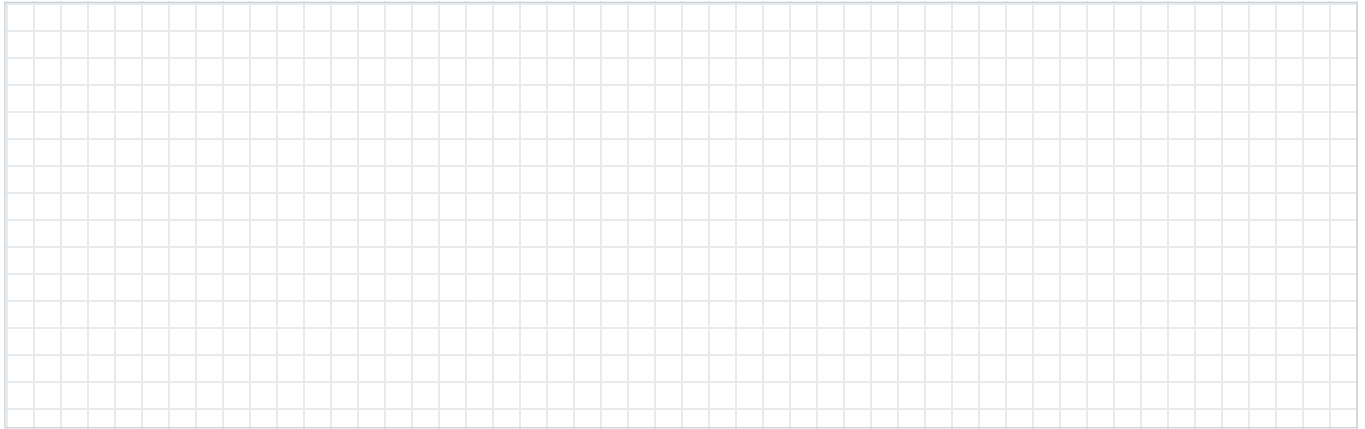
Gear ratio

Power

Floor

Program version

Sketch Bevel Bot. Mark its motor and the edge you measured from.



Optional: try one change

One change / reason	Travel (cm) - run 1	Travel (cm) - run 2	Travel (cm) - run 3	Mean (cm)	Range (cm)
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Core check: what did you build, and did it drive and stop?

My contribution / next question

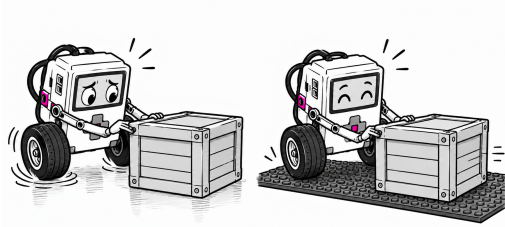
One thing I can do now...

Core complete? Choose one (optional): Try a second distance OR tune one setting and repeat three runs.

Find your grip



Mission: Find the setup that slides at the steepest angle.



"My wheels spin. The crate stays put!"

Grip depends on the tires and the surface.

Key idea

Friction — A force that opposes sliding, or the tendency to slide, between two surfaces in contact. Example: Friction resists a book sliding across a table.

Brace — A support that helps a structure resist changing shape. Example: A diagonal beam stiffens the robot's frame.

On our robot, tire-floor friction provides grip.

Predict first

Which tire or surface will slide at a steeper angle?

Try it

- 1 Use Bevel Bot. Ask a mentor to check the ramp and wheel restraint.
- 2 Raise slowly. Record the angle at first sliding, three times.
- 3 Keep mass and setup fixed; change only the tire or surface.
- 4 Add a brace; repeat a gentle push. See the facing page.

Keep the evidence

Setup	Slide 1 (°)	Slide 2 (°)	Slide 3 (°)	Mean (°)	Range (°)
-------	-------------	-------------	-------------	----------	-----------

A

B

Rolling or tipping is a different result. Record it; reset the test.

One thing I noticed

Beyond sumo • Why do a wheelchair and a race car need different tires?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Surface _____

Tires _____

Fixed mass (g) _____

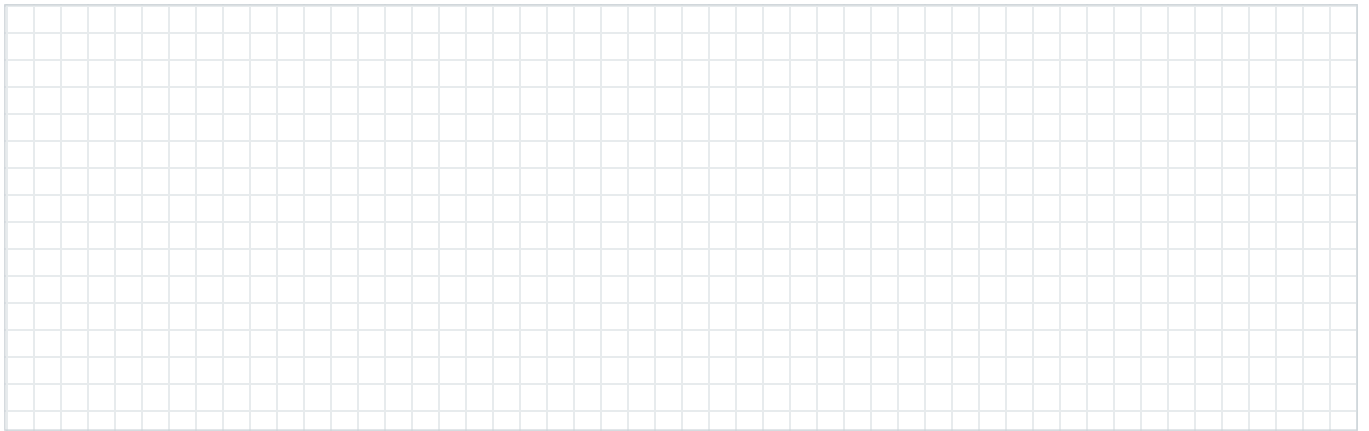
Wheel restraint _____



Grip test: hold wheels still. Read the angle at first slide.

Brace test: add a diagonal. Repeat the same gentle push.

Sketch the weak connection and the brace you added.



Check the brace

Same gentle test

Before brace

After brace

Did it resist the same gentle push better?

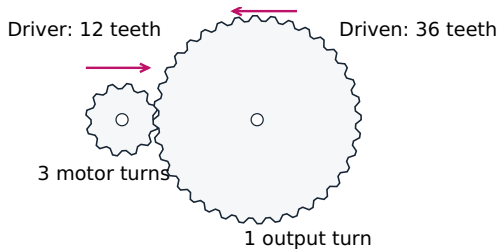
My contribution / next question

One thing I can do now...

Speed or shove?



Mission: Compare three gear setups. Pick your trade-off.



Key idea

Gear — A toothed wheel that meshes with another gear to transmit rotation. Example: One gear turns another.

Torque — The turning effect of a force about an axis. Example: Pushing a wrench turns a nut.

Gear ratio — For one gear pair: driven teeth $\div$ driving teeth. Example: $36 \div 12 = 3:1$ reduction.

Predict first

Which will be fastest? Which will pull best?

Try it

- 1 Test one assigned prepared gear station with a mentor. Set its timed drive for the 2 m lane; record three runs.
- 2 Copy the other two stations' results; name their data source. Keep your Bevel Bot intact for today.
- 3 For the mentor-led tug test: drive for 0.5 seconds, then stop. Swap sides and repeat.

Keep the evidence

Ratio	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Mean (s)	Range (s)
1:1					
3:1 reduction					
1:3 increase					

Different robots can differ in more than gearing. Note those differences.

One thing I noticed

Beyond sumo • Why does a bicycle have more than one gear?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Station / robot

Data source

Wheels

Power

Floor

Sketch the driver and driven gears; label tooth counts.

Tug-test evidence

Pair	First result	Sides swapped	Slip / stall / drag
A vs B			
Other (optional)			

What did you gain? What did you give up?

My contribution / next question

One thing I can do now...

Core complete? Choose one (optional): Calculate speed in cm/s OR compare another short tug pair with a mentor.

Choose your fighter



Mission: Build a two-motor base. Choose one useful improvement.

A choice has a gain and a cost.

REDUCTION GEARS

More torque
Less speed

SPEED-UP GEARS

More speed
Less torque

Which matters for your mission?

Key idea

Constraint — A requirement or limit a design must satisfy. Example: Fit inside 30 × 30 cm.

Trade-off — Gaining one benefit while giving up another. Example: More bracing adds stiffness but also mass.

Prototype — A model or early version built to test an idea. Example: Build a trial sensor mount.

Predict first

Which feature will help most? What might it cost?

Try it

- 1 Save a Bevel Bot photo and program. Sketch two small improvements for your next robot.
- 2 [Follow LEGO's two-motor Driving Base pictures](#). Reuse the EV3 kit. Connect the drive motors to B/C.
- 3 Keep the standard drivetrain. Check forward/stop (Programming Guide p. 2), size, and room for sensors. Add one improvement if time remains.

Keep the evidence

Choice	Our pick	Evidence / trade-off
Gears		
Wheels		
Height / balance		

Start inside 30 × 30 cm. Leave space for the sensors. Rules: pages 30-31.

One thing I noticed

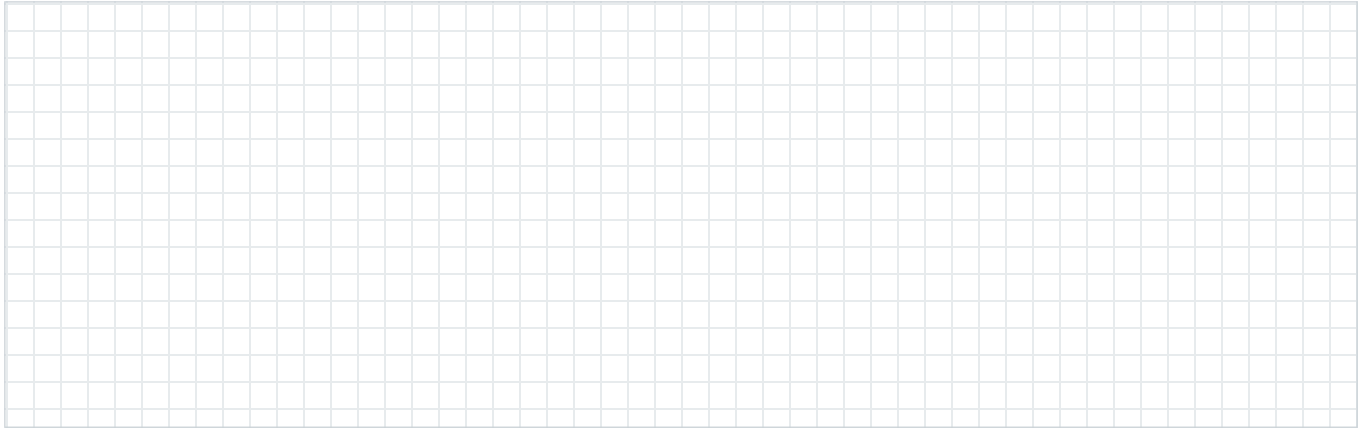
Beyond sumo • What might a rescue robot need that a sumo robot does not?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Our top priority

Two ideas - label the parts, dimensions, and sensor positions.



Our design choice

Chosen idea	Why this one?	What we gave up
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Which assumption did we test? What evidence convinced us?

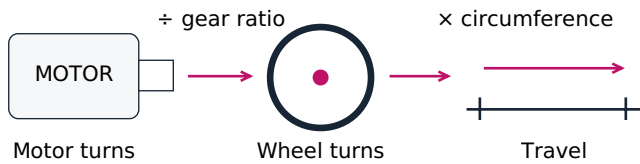
My contribution / next question

One thing I can do now...

Make it move



Mission: Turn a motor command into a measured journey.



Key idea

Circumference — The distance around a circle. Example: A 6 cm wheel has circumference $\pi \times 6 \approx 18.85$ cm.

Motor rotation — A turn of the motor axle; one full rotation is 360° . Example: Two full rotations equal 720° .

Predict first

How many motor turns will move our robot 100 cm?

Try it

- 1 Use the motion card on page 32 to calculate a starting command.
- 2 Use the B/C Movement stack in Programming Guide p. 2 with your calculated motor turns. Save and download it.
- 3 Measure three unchanged runs. Compare prediction with travel.

Keep the evidence

Trial	Actual travel (cm)	Error (cm): actual - 100
1		
2		
3		
Mean		
Range		

Motor turns and wheel turns differ when gearing changes.

One thing I noticed

Beyond sumo • How could a warehouse robot deliver to the right shelf?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Diameter (cm)

Circumference (cm)

Gear ratio

Motor turns

Power

Program version

Draw the program or write its steps. Label settings and ports.

A second setting

One change / reason	Travel (cm) - run 1	Travel (cm) - run 2	Travel (cm) - run 3	Mean (cm)	Range (cm)
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Why did measured travel differ from the prediction?

My contribution / next question

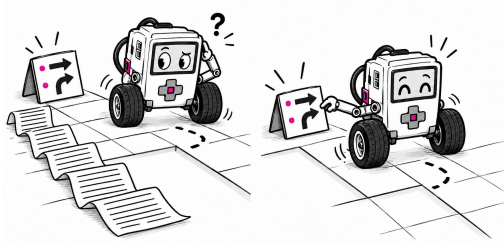
One thing I can do now...

Core complete? Choose one (optional): Predict a new distance OR tune one setting for the same target.

Repeat and turn



Mission: Use a loop to drive a square.



A long list repeats the same instructions.

A loop reuses DRIVE + TURN.
For a square: repeat 4 times.

Key idea

Loop — A programming structure that repeats instructions. Example: Repeat DRIVE + TURN four times.

Heading — The direction an object faces. Example: A quarter-turn changes heading by 90°.

Predict first

What heading change would a triangle need? Guess before page 33.

Try it

- 1 Build a turn (Programming Guide, p. 3). Tune for 90°; confirm with three turns.
- 2 Loop: drive one side, turn. Repeat four times.
- 3 Core: run one square. If time remains, complete three unchanged square trials; then calculate mean/range.

Keep the evidence

Square run	Finish from start (cm)	Heading error (°)
1		
2		
3		
Mean		
Range		

A triangle is an extension. Robot heading degrees are not motor degrees.

One thing I noticed

Beyond sumo • Where would repeating a route help a robot do its job?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Side length (cm)

Motor command (°)

Turn type

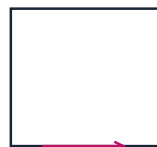
Power

Program version

Draw your loop or path. Mark where each turn happens.

Turn calibration

Motor command (°)	Heading 1 (°)	Heading 2 (°)	Heading 3 (°)	Mean (°)	Range (°)
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4 sides

Repeat 4 times

Drive one side



Turn 90°

Where did small errors add up?

My contribution / next question

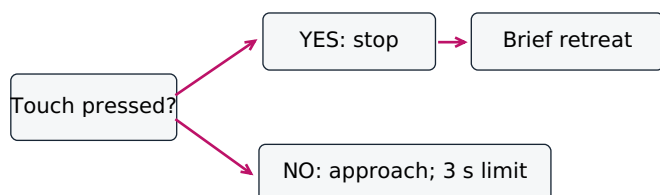
One thing I can do now...

Core complete? Choose one (optional): Predict and build a triangle OR a hexagon. Calibrate its turn first.

Feel a hit



Mission: Detect contact. Stop and back away once.



Key idea

Sensor — A device that detects or measures a physical quantity or condition. Example: A touch sensor detects contact.

Condition — An expression that evaluates to true or false. Example: Is the touch sensor pressed?

Timeout — A time limit that ends an operation when reached. Example: Stop the approach after 3 seconds.

Predict first

How far from the wall will the front bumper finish?

Try it

- 1 Mount touch (Guide p. 4), port 1. Motors off: repeat IF pressed, beep; ELSE quiet.
- 2 Build a loop: approach slowly; check touch and time. Stop on contact or after 3 seconds.
- 3 Reverse only after contact. Extra: three gap trials.

Keep the evidence

Trial	Final gap (cm)	Contact or timeout?
1		
2		
3		
Mean		—
Range		—

Gap summaries: use contact trials only; exclude timeouts. Record how many you used in your notes.

Use a stable barrier and clear rear space. This is a contact exercise, not the final match strategy.

One thing I noticed

Beyond sumo • What should an automatic door do when something blocks it?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

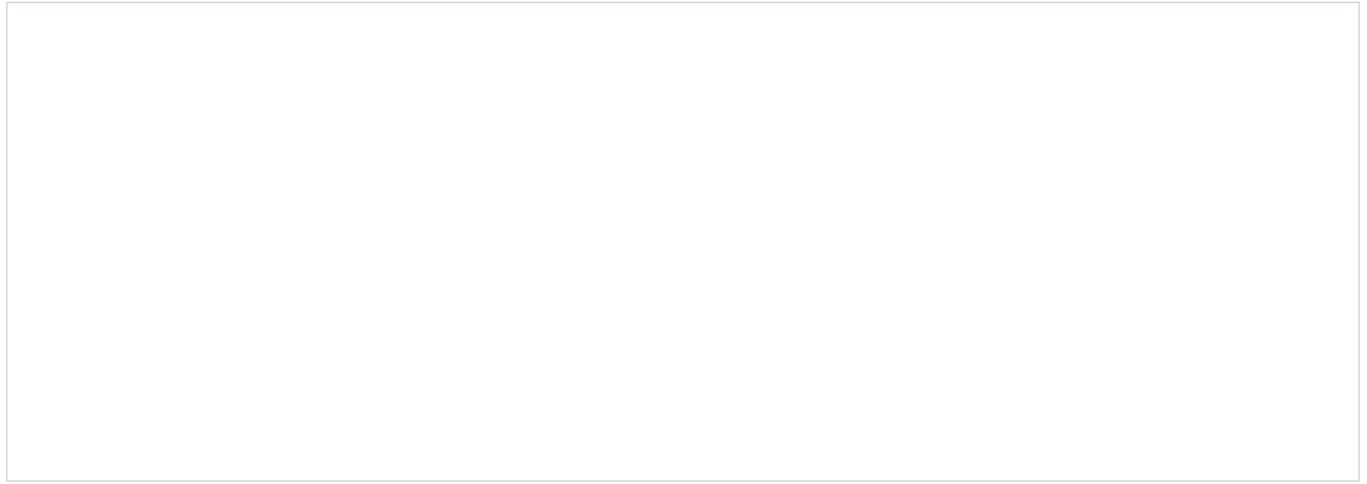
Approach power

Reverse time (s)

Target gap (cm)

Program version

Draw the touch condition and both outcomes: contact / timeout.

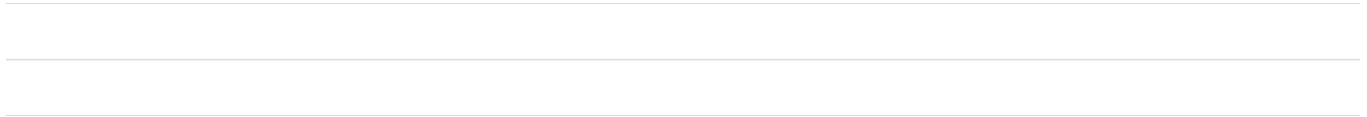


Change one setting

One change / reason	Gap (cm) / timeout - run 1	Gap (cm) / timeout - run 2	Gap (cm) / timeout - run 3	Mean (cm)	Range (cm)
------------------------	-------------------------------	-------------------------------	-------------------------------	-----------	------------

Gap summaries: use contact trials only; exclude timeouts. Record how many you used in your notes.

What happened when no contact was detected?



My contribution / next question

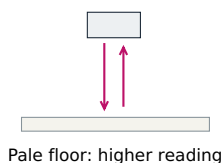
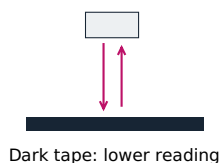
One thing I can do now...



Can it stay in?



Mission: Stay inside the ring for 60 seconds.



Key idea

Reflected light — Light returned from a surface.

Example: The floor reflects the sensor's light.

Threshold — A cutoff value used to make a decision.

Example: If light < 35, begin edge recovery.

Calibration — Establishing how instrument readings relate to known reference values. Example: Relate distance-sensor readings to ruler distances.

Here, floor/tape measurements help set a threshold.

Predict first

What reading should mean “edge”?

Try it

- 1 Mount downward light sensor (Guide p. 5), cable to 3. Read floor and tape five times at the same height.
- 2 Choose a cutoff between the groups. Build a loop: if edge, stop, reverse briefly, then turn; otherwise drive slowly.
- 3 After recovery, read again; still at the edge? Stop. End each run at 60 seconds. Core: one recovery. Extra: three trials; log end reasons opposite.

Keep the evidence

Surface	1	2	3	4	5	Mean	Range
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Tape

Floor

Readings overlap? Fix the setup first. A front sensor cannot see behind or beside the robot.

One thing I noticed

Beyond sumo • How could a robot vacuum avoid a stair edge?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Light cutoff

Compare: < or >

Sensor height (cm)

Power

Program version

Draw the edge program or sensor mount. Label what matters.

Three boundary trials

Trial	Elapsed run (s)	End reason
1		
2		
3		
Mean		—
Range		—

End reason: 60 s / exit / recovery stop / mentor stop. A stopped robot inside the ring has not demonstrated successful recovery. Compare times only for the same test and end reason.

What one change would you test next?

My contribution / next question

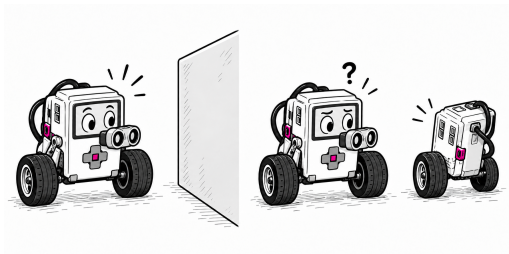
One thing I can do now...

Optional with a mentor: reverse near an edge OR try a gentle side push (p. 37).

Find the opponent



Mission: Find a target with sound. Keep motors off today.



"That distance could be a wall!"

An angled target may return a weak echo.

Predict first

At what distance should we say "target nearby"?

Keep the evidence

Actual distance	Reading 1 (cm)	Reading 2 (cm)	Reading 3 (cm)	Mean (cm)	Range (cm)
10 cm					
30 cm					
60 cm					

Record no echo/errors as outcomes, not distances. Average valid distance readings only; write the count used. No echo is not proof of a fault.

One thing I noticed

Key idea

Ultrasonic distance sensor — A device that measures distance by timing returning high-frequency sound echoes. Example: It estimates the gap to a wall.

Echo — Sound reflected back from a surface. Example: A sound pulse returns from the wall.

Try it

- 1 Mount distance sensor (Guide p. 5), port 4. Loop: read/display cm, wait 0.1 s. From sensor face, test 10/30/60 cm; three readings each.
- 2 Test an angled target and the empty background.
- 3 IF a usable reading is nearby, beep; ELSE stay quiet.

Beyond sumo • How could a parking sensor warn a driver about a nearby object?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

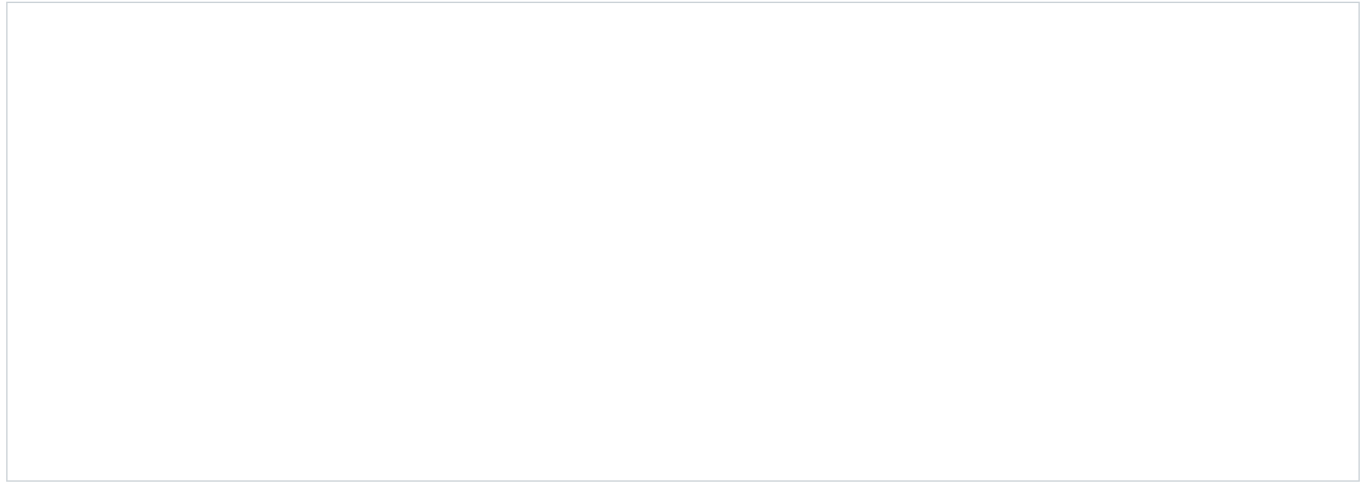
Sensor height (cm)

Target angle (°)

Near interval (cm)

Program version

Sketch the sensor view and the target that fooled it.



Detection clues

Case	Displayed reading / indication	Our conclusion
Angled target		
No echo		
Disconnected: motors off		

Which assumption about sensing did our test challenge?

My contribution / next question

One thing I can do now...

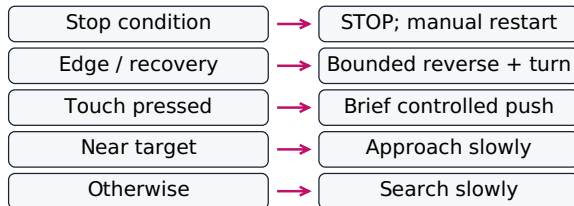
Core complete? Choose one (optional): Motors off: test 90 cm OR compare two target angles.

Choose what happens



Mission: Combine your robot's decisions in one loop.

Check top to bottom. First matching rule wins.



Key idea

Priority — The order of precedence among actions or rules. Example: Handle an edge before approaching a target.

State — The current mode or stage of a system. Example: The robot is in REVERSE mode.

Timer — A mechanism that measures elapsed time. Example: Stop when match time reaches 60 seconds.

Explain the order

Why should edge recovery come before approaching an opponent?

Try it

- 1 Build in guided stages (Programming Guide pp. 7-10). Reuse your tested settings. Trace one branch at a time with your mentor.
- 2 Use short commands and repeated sensor checks.
- 3 Test conflicts, no echo, contact limits, and recovery. Record retests.

Keep the evidence

Case	Expected	Actual
Edge + target		
Touch; floor clear		
No confirmed target		

Fault → stop. Edge → bounded recovery. Contact → brief push, not automatic retreat.

One thing I noticed

Beyond sumo · What should a delivery robot do if its path is blocked near a curb?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Edge cutoff

Near interval (cm)

Power

Program version

Draw your decision flow. Show where it loops back.

Test the awkward cases

Case	Expected	Actual / pass?	Change / retest
Edge during approach	Start recovery		
Touch held ≥ 3 s	STOP		
Detectable fault	STOP		
Dark after recovery	STOP		

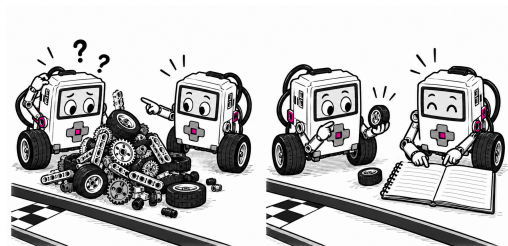
Which change fixed a failing case?

My contribution / next question

One thing I can do now...

Make one improvement

Mission: Make one change. Show whether it helped.



"I changed everything.
What helped?"

Test 3 times. Change one
thing.
Test 3 more times.

Key idea

Experimental variable — A factor that can change or be measured in a test.
Example: Change wheel size; measure travel.

Mean — The sum of values divided by their count. Example: 98, 99, 100 cm: mean = 99 cm.

Range — The largest value minus the smallest. Example: 98, 99, 100 cm: range = 2 cm.

Predict first

What will improve? What might get worse?

Try it

- 1 Choose one metric and a target. Record three baseline runs.
- 2 Change one thing. Repeat the same test three times.
- 3 Compare both means with your target and both ranges. Keep or undo the change; explain why.

Keep the evidence

Version	Trial 1	Trial 2	Trial 3	Mean	Range
Before					
After					

Record metric/unit opposite; use that unit for trials, mean, and range. Check the starting size again.

One thing I noticed

Beyond sumo • How could you tell whether a change improved a paper airplane?

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

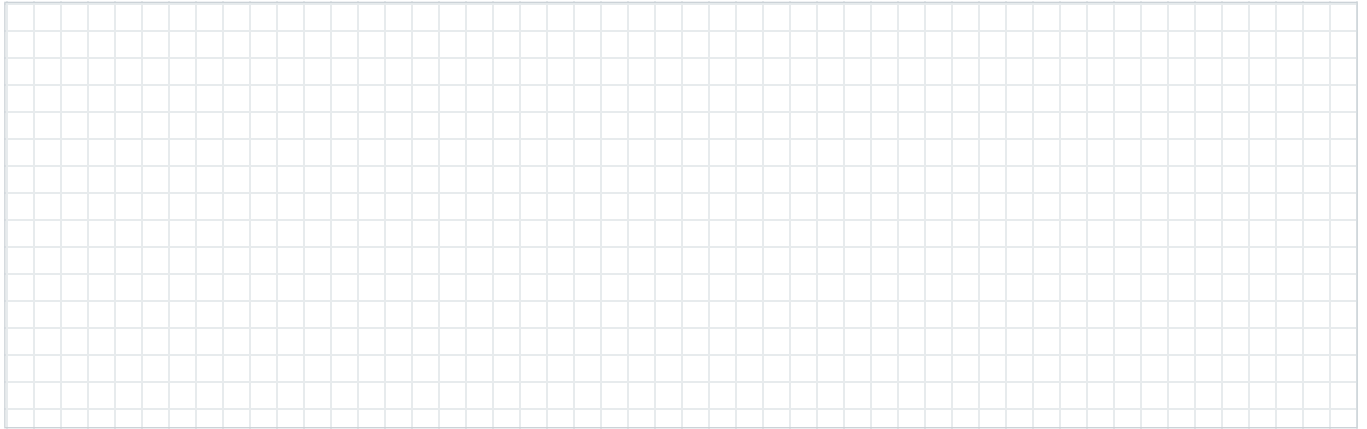
Metric

Unit

Success target

Fixed conditions

Sketch the change. Label the new part, position, or setting.



Our decision

Keep / undo

What improved?

What got worse?

What did we assume would improve? Did the evidence agree?

My contribution / next question

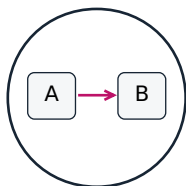
One thing I can do now...

Core complete? Choose one (optional): Ask another team to repeat your test OR repeat it on a second surface.

Tournament day



Mission: Welcome your family. Compete, learn, and explain your design.



Explain your design, too.

3 points: win

1 point: draw

0 points: loss

Key idea

Evidence — Observations or measurements used to support or evaluate a claim. Example: Three tests showed less slipping.

Iteration — One repetition of a process used to develop or improve something. Example: Revise the design, test, and review again.

Predict first

Which of our design choices will matter most?

Try it

- 1 Check battery, cables, size, sensors, and saved program.
- 2 Use the marked starts and 5-second wait. Follow the referee.
- 3 Record results and congratulate each opponent.

Keep the evidence

Opponent	W / D / L	Points	What happened?

Win 3 • draw 1 • loss 0. Points are compared within your completed group.

One thing I noticed

Our ideas and evidence

Actual date: _____ My role: _____ Program/version: _____

Final program

Gear ratio

Edge cutoff

Sketch the design feature you most want to explain.

Make your design case

Evidence	Trade-off	Tested improvement
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What did each teammate contribute?

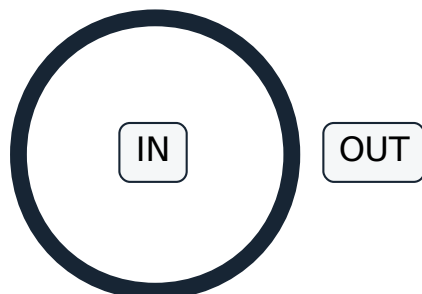
My contribution / next question

One thing I can do now...

Beyond sumo · Which of your robot skills could help solve a problem at school?

Sumo: know the rules

Compete hard. Treat people and robots well.



Outer tape edge = boundary

Main-body contact points count

Detached parts do not keep you in

Your robot

Start inside **30 × 30 cm**.
Safe LEGO parts from your kit.
Runs on its own.

Your match

Flat pale floor; 120 cm outer-diameter ring,
5 cm dark tape. Marked opposite starts.
5-second wait, then 60 seconds.
No hands during play.

Outcome	Points
Win: opponent out first	3
Draw: neither out or simultaneous exits	1
Loss	0

No damaging parts, sticky surfaces, projectiles, grabbing, trapping, or tying. Remove the tug-test link.

A robot is out when all main-body ground contacts are beyond the outer tape edge. Touching tape remains in. Design awards recognize evidence, trade-offs, and tested improvement.

When a match gets messy

Situation	What happens
Early start	Reset once. A second early start by that team in the match is a loss.
Uninvited touch	First incident: reset and warning. Second by that team in the match: loss.
Tipped robot	Up to 10 seconds to self-right within the match clock; otherwise loss. Both fail together: draw.
Stall or possible damage	Referee stops promptly. Draw if no result was already decided; inspect before the next match.
Loose part	Main body determines in/out. Leave a safe piece; stop and restart after repair if it traps a robot or creates danger.
External interruption	Stop and restart from the marked positions. Referee resolves close calls. A program STOP stays stopped until the referee ends or resets the match; no mid-match manual restart.
Standings	Complete round-robin groups. Recognize pool winners; do not compare raw points across pools.
Tie for placing	Head-to-head only if it resolves the tie; otherwise share placing.

Design award: 0-2 each for evidence, trade-off explanation, and tested improvement. Tied scores share recognition.

One rule I want to clarify before we start

Pocket ideas: measure and move

Mean and range

98, 99, 100 cm

Mean = **99 cm**

Range = **2 cm**

Add valid readings; divide by their count. Range = biggest – smallest. Three trials are a sample, not a guarantee.

Error from the target

Actual – target

98 – 100 = **–2 cm**

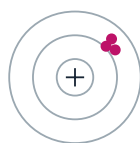
Negative: short.

Positive: beyond.

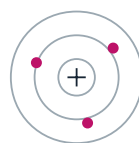
Precision: runs agree. Accuracy: close to target.



Close + consistent



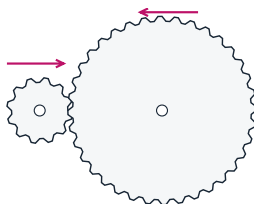
Consistent, off target



Scattered results

Systematic error: a consistent bias. Random error: trial-to-trial variation. Both can occur together.

Gears trade speed for torque



12-tooth driver: 3 motor turns.

36-tooth driven gear: 1 output turn.

Driver → driven	Ratio: driven ÷ driver	Ideal output
12 → 36	3:1 reduction	$\frac{1}{3}$ speed; 3× torque
24 → 24	1:1	Same speed/torque
36 → 12	1:3 increase	3× speed; $\frac{1}{3}$ torque

Torque means turning effort at an axle. Real gears lose energy; pushing also depends on grip and wheel size.

Predict travel

Wheel turns = motor turns ÷ ratio

Circumference = $\pi \times$ wheel diameter

Travel = wheel turns $\times$ circumference

Same units throughout. Assumes no slip; test the prediction.

Pocket ideas: turn and sense

Turn the robot

Square: $4 \times 90^\circ$

Triangle: $3 \times 120^\circ$

Heading change = $360^\circ \div \text{sides}$. Motor degrees need calibration.

Touch is a switch



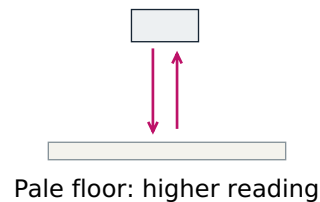
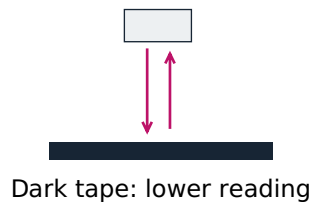
Released



Pressed

No force or direction measurement. Check repeatedly to notice a change.

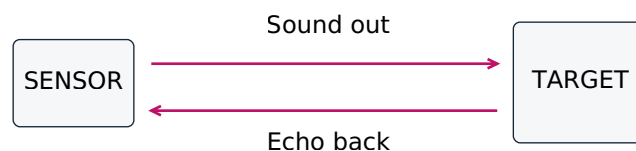
Reflected light



Measure at the mounted height. If tape and floor groups are separate, choose a threshold between them. Recheck when conditions change.

A front sensor cannot watch the rear or side edge.

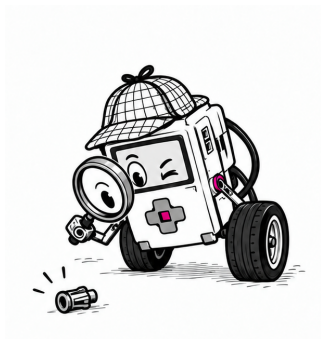
Distance is a clue



A distance is a clue, not an identity.

No echo: no confirmed target. Detected read/connection fault: stop. A plausible wrong reading may go unnoticed.

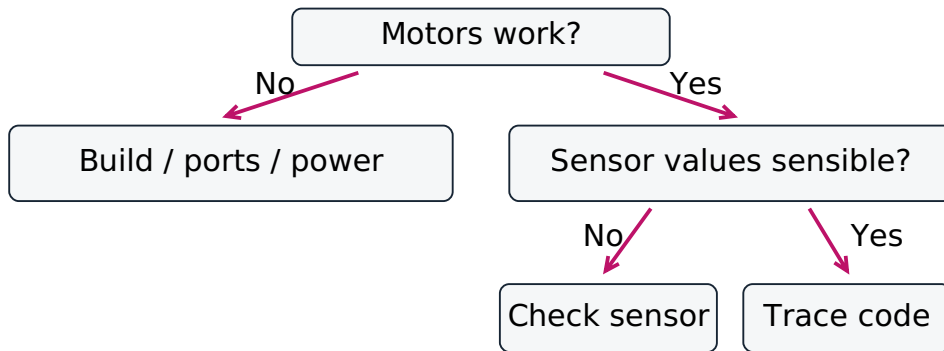
Be the robot detective



Start with one clue.

What did you expect?

What actually happened?



Clear lane, low speed for the motor check; motors off for sensor checks. After one change, repeat the failing test.

Check	A small test
Basics	Stop. Battery, ports, cables, latest download.
Build	With power stopped: loose, rubbing, bent, blocked?
Sensor	Motors off: display readings for known situations.
Code	Test a short section. Add a beep or display to locate the fault.

Change one thing where possible. Repeat the original failing test. Keep the earlier result so you can compare.

Our clue → our test → what we learned

Show what you can do

H = with help · I = independently · E = independently and can explain. You can try again.

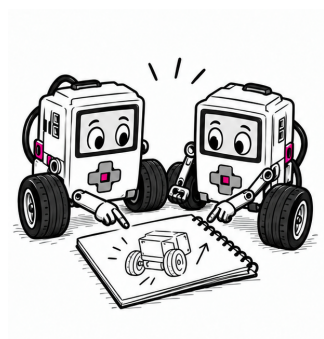
Week	Show a mentor	H/I/E	Date
1	Build Bevel Bot; drive and stop safely.		
3	Explain gearing and a traction limit.		
5	Connect motor turns to measured travel.		
6	Explain a loop and a calibrated turn.		
7	Show a condition and a timeout.		
8	Choose a threshold; show a blind spot.		
9	Explain a missed or false detection.		
10	Show priorities, contact limit, and fault response.		
11	Use before/after evidence to defend a change.		

Something I can do now / something I want to practice

Missed a week? Rejoin here.

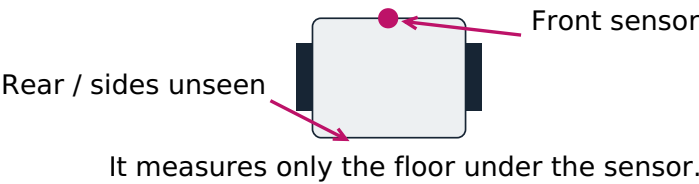
You do not have to start over.

Week	Small bridge back into the team	Pages
1	Locate Bevel Bot's motor; trace p. 5; explain a drive/stop test.	6-7
2	Observe grip; locate a brace.	8-9
3	Read station data; explain our ratio.	10-11
4	Identify B/C motors on the new base; show forward/stop.	12-13
5	Open the saved program; predict a run.	14-15
6	Trace four repeats; show one turn.	16-17
7	Check touch mount/cable; run a motors-off touch check.	18-19
8	Check downward mount/port 3; remeasure floor/tape; trace loop.	20-21
9	Check forward mount/port 4; read target/background, motors off.	22-23
10	Trace the team's decisions; show edge + target priority.	24-25
11	Read before/after results; find final version.	26-27
12	Explain your design; reflect.	28-29



What I caught up on / what help I need

Test the blind spots



Low speed. Clear floor. Hands clear of moving parts. Record the setup before testing.

Test	Expected	Actual	Retest
Reverse near edge: rear clearance + reverse time			
Gentle side push: side clearance + sensor path			
No echo / background			
Disconnected sensor: motors-off diagnosis			
Touch held: automatic stop limit			

No echo does not prove a fault. Do not claim a missing-sensor stop unless your app exposes it and you have demonstrated it.

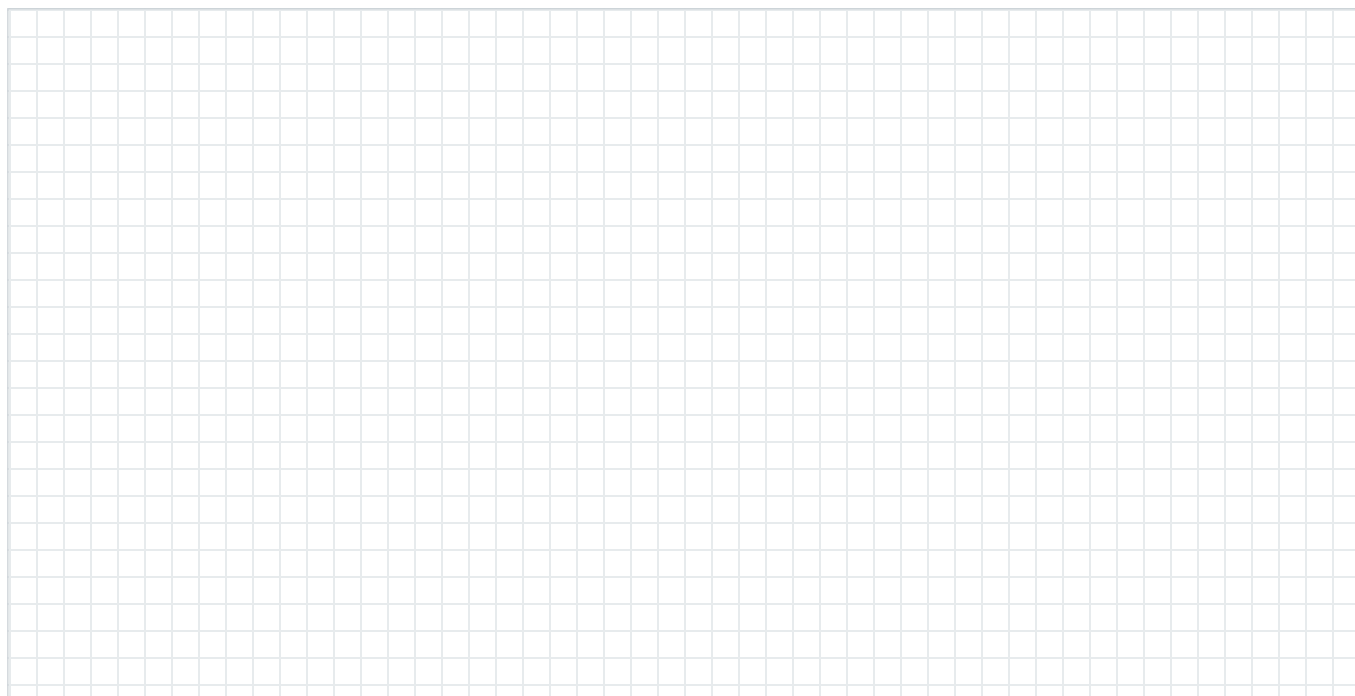
Extra setup / unit	Trial 1	Trial 2	Trial 3
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What still gives our robot trouble?

A page for your own idea

Problem / goal / limits

Sketch two ways to solve it



Which will I try? Why?

What happened? What will I change?

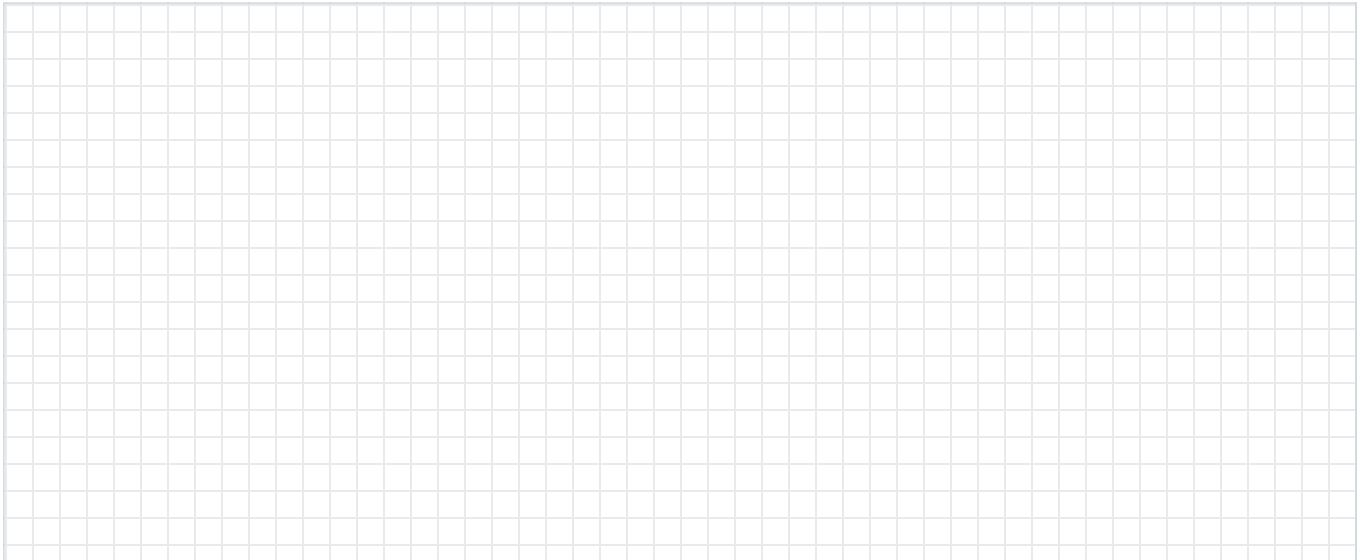
Engineering beyond the ring

Who could your next idea help?

What happens now? What would be better?

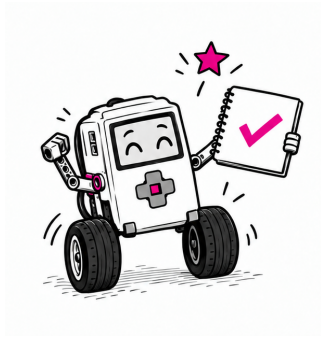
What should it do - and how will I know?

Sketch a small first version



How I would test it / two limits to work within

Look what you figured out



You have evidence of your progress.

Keep this book. Your next idea can start here.

Look back at page 3. Did you learn what you wanted to learn?

What can I do now? Point to a page.

Which surprise taught me the most?

How did I help? Who helped me?

What do I want to build next?
